# Dual-State KV Forking: Low-Overhead Semantic Interruption for Full-Duplex LLM Agents

**Anonymous Author(s)**
Affiliation
Address
email

## Abstract

Human conversation is full-duplex: listeners process speech continuously and interrupt mid-utterance to correct errors, volunteer answers, or take the floor. Multi-agent LLM systems remain half-duplex: a listener must wait for the speaker's entire turn, and input streaming—while it keeps the listener's cache warm—remains passive. We present the **Dual-State KV-Forking** architecture, which gives a streaming listener *active* floor control. The listener maintains a progressively updated Base KV cache over the incoming stream; at each chunk boundary it forks this cache (0.01–0.06 ms measured) into an ephemeral Assessor branch that appends a short decider prompt and emits a single logit-gated `STOP`/`CONTINUE` token, leaving the Base state intact. Each stream token is ingested exactly once, eliminating the $O(N^2 \cdot k)$ context re-prefill of a stateless independent decider while preserving a warm cache for instant response generation. Across three models on physical hardware, on FLEXI, Full-Duplex-Bench, and a 200-scenario progressive-clue floor-control task, our framework matches or exceeds independent-decider accuracy (up to 100% on FDB), cuts total tokens processed per scenario by **78.6%**, reduces Time-to-Halt by up to **49.1%** under concurrent full-duplex streaming, and cuts no-interruption Time-to-First-Token by up to **65.0%**.

## 1 Introduction

Human dialogue is fluid and full-duplex: listeners emit backchannels ("uh-huh", "right") to signal comprehension without claiming the floor, and execute intentional barge-ins ("wait, stop") when they detect an error, possess the answer, or must inject a constraint (Sacks et al., 1974; Yngve, 1970; Skantze, 2021). LLM multi-agent systems (Wu et al., 2023; Li et al., 2023), however, are half-duplex: an agent waits passively for the speaker to complete its turn, forcing the speaker into an asymmetric dilemma—verbosity wastes tokens, latency, and context; brevity risks ambiguity—and leaving the listener powerless against hallucination, redundant tool calls, or obsolete instructions mid-stream. *Input streaming* (Xiao et al., 2024; Agrawal et al., 2024) partially addresses end-of-turn latency by prefilling tokens as they arrive, minimizing Time-to-First-Token (TTFT) once the speaker finishes—but it is passive; the listener still cannot seize the floor. What is missing is **semantic interruption assessment**: continuously classifying the incoming stream as *intentional barge-in* (`STOP`) versus *benign continuation* (`CONTINUE`).

The naive solution—invoking an independent LLM decider on every streaming chunk—introduces three bottlenecks. **(1) Quadratic re-prefill**: a stateless decider re-prefills the accumulated prefix of $m \cdot k$ tokens at every check $m$, a cumulative $\sum_{m=1}^{N} mk = O(N^2 \cdot k)$ prefill cost that dwarfs the dialogue itself in long interactions. **(2) Reasoning latency**: instruction-tuned deciders emit 15–35 chain-of-thought tokens before answering, adding seconds of delay to a millisecond-scale decision. **(3) Cold-start TTFT**: if no interruption fires, the decider's transient context is discarded and the listener must re-prefill the full $N \cdot k$ sequence before responding.

Submitted to 40th Conference on Neural Information Processing Systems (NeurIPS 2026). Do not distribute.

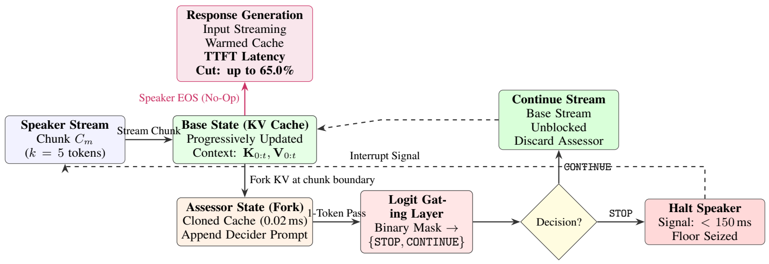


Figure 1: **Dual-State KV-Forking.** Speaker chunks update the Base cache ($O(k)$/chunk). At each boundary the cache is forked ($\leq$0.06 ms) into an Assessor emitting one gated STOP/CONTINUE token. CONTINUE discards the fork; STOP halts the speaker in sub-150 ms; on natural completion the warm cache generates the response immediately.

We introduce the **Dual-State KV-Forking** architecture (Figure 1). A *Base state* ingests the stream once, chunk by chunk, into a persistent KV cache. At each chunk boundary, the cache is forked—via shallow tensor cloning, measured at 0.01–0.06 ms—into an ephemeral *Assessor state* that appends a short decider prompt and emits a single token constrained by logit gating to $\{\texttt{STOP}, \texttt{CONTINUE}\}$. The fork is discarded after the decision; the Base cache is never touched, so it remains warm for instant response generation if the speaker concludes naturally.

Our contributions are: **(i)** a dual-state execution scheme adding active, prompt-configurable interruption to streaming listeners with per-check prefill cost independent of history length, zero trained parameters, and no corruption of the response-generation state; **(ii)** a three-model physical-hardware study spanning interruption accuracy on FLEXI and Full-Duplex-Bench (Lin et al., 2025) and end-to-end floor control on a 200-scenario progressive-clue task built on incremental QA (Rodriguez et al., 2019; Wallace et al., 2019); **(iii)** accuracy on par with (or better than) a reasoning independent decider at **78.6%** fewer total tokens, up to **49.1%** lower Time-to-Halt, and up to **65.0%** lower no-op TTFT via the preserved streaming cache.

## 2 Related Work

**Full-duplex spoken dialogue.** Turn-taking, overlap, and backchannels are foundational in conversation analysis (Sacks et al., 1974; Yngve, 1970) and long engineered into spoken dialogue systems (Skantze, 2021; Lin et al., 2022). Neural full-duplex models jointly listen and speak—dGSLM (Nguyen et al., 2023), Moshi (Défossez et al., 2024), duplex-tuned LLMs (Zhang et al., 2024)—and Full-Duplex-Bench (Lin et al., 2025) standardizes evaluation of pause handling, backchanneling, turn-taking, and interruptions. These works target the speech interface of a *single* agent facing a human; we target the complementary problem of *agent-to-agent* floor control in text-token streams with a frozen LLM listener.

**Streaming ingestion and KV reuse.** Chunked prefill (Agrawal et al., 2024), attention sinks (Xiao et al., 2024), continuous batching (Yu et al., 2022), paged KV memory (Kwon et al., 2023), radix-tree prefix sharing (Zheng et al., 2024), prompt caching (Gim et al., 2024), and speculative decoding (Leviathan et al., 2023) all reuse or cheaply extend computed state to cut latency and memory. Our contribution is not a new cache mechanism but a *dual-state execution pattern* over these primitives—a persistent streaming trunk plus ephemeral copy-on-write assessment branches—applied to real-time interruption; it maps natively onto RadixAttention/PagedAttention (Appendix F).

**Multi-agent LLM systems and incremental QA.** Multi-agent frameworks such as AutoGen (Wu et al., 2023) and CAMEL (Li et al., 2023) coordinate agents through strictly sequential message exchange. Incremental QA—Quizbowl (Rodriguez et al., 2019) and adversarially authored questions (Wallace et al., 2019)—provides a natural testbed for *when to buzz*, which we adapt as a full-duplex floor-control benchmark.

## 3 The Dual-State KV-Forking Framework

The framework decomposes stream comprehension into two decoupled execution states (Figure 1).

Table 1: **Interruption accuracy and latency.** Mean decision latency on FDB; Time-to-Halt on true positives. Full tables with confusion matrices and FLEXI latencies in Appendix A.

| Model | Architecture | FLEXI Acc. | FDB Acc. | Avg. FDB Latency | Time-to-Halt (FDB TP) |
|---|---|---|---|---|---|
| **Gemma 4 12B** | Independent (reasoning) | **99.75%** | 99.50% | 2452.00 ms | 1117.61 ms |
| | Independent + Logit Gating | 98.75% | 90.75% | 266.06 ms | 268.46 ms |
| | **Dual-State (ours)** | 98.75% | **100.00%** | **90.26 ms** | **180.53 ms** |
| **Llama 3.1 8B** | Independent (reasoning) | **93.50%** | **97.00%** | 361.96 ms | 518.45 ms |
| | Independent + Logit Gating | 91.00% | 95.00% | 117.07 ms | 117.67 ms |
| | **Dual-State (ours)** | 90.50% | 94.25% | **57.87 ms** | **115.73 ms** |
| **Qwen 3 4B** | Independent (reasoning) | 97.50% | **99.75%** | 189.64 ms | 191.19 ms |
| | Independent + Logit Gating | 98.75% | **99.75%** | 122.35 ms | 122.63 ms |
| | **Dual-State (ours)** | **99.00%** | 99.50% | **42.42 ms** | **85.70 ms** |

**Base state: continuous streaming ingestion.** Let $S = (x_1, \ldots, x_T)$ be the speaker's token sequence, arriving in chunks $C_m = (x_{(m-1)k+1}, \ldots, x_{mk})$, $m \in \{1, \ldots, M\}$. On receiving $C_m$, the Base state computes projections only for the new tokens ($\mathbf{K}_{\text{new}}^{(m)} = C_m \mathbf{W}_K$, $\mathbf{V}_{\text{new}}^{(m)} = C_m \mathbf{W}_V$), appends them to the cache, $\mathbf{K}_{\text{base}}^{(m)} = [\mathbf{K}_{\text{base}}^{(m-1)} \| \mathbf{K}_{\text{new}}^{(m)}]$ (analogously for $\mathbf{V}$), and attends $\mathbf{Q}_{\text{new}}^{(m)}$ causally over the accumulated cache. Past KV states persist, so each stream token is ingested exactly once: cumulative prefill cost is linear, $\mathcal{T}_{\text{base}} = \sum_{m=1}^{M} k = T$.

**Assessor state: low-overhead KV forking.** To evaluate an interruption at chunk $m$, the system forks the Base cache, $(\mathbf{K}, \mathbf{V})_{\text{assessor}}^{(m)} \leftarrow (\mathbf{K}, \mathbf{V})_{\text{base}}^{(m)}$, via pointer aliasing (measured as 0.01–0.06 ms deep-clone overhead in our PyTorch validation prototype, and zero-copy in production engines; Appendix F). The fork isolates the assessment: the decider prompt never contaminates the Base cache, which continues ingesting the live stream.

**Logit gating.** The Assessor appends a fixed decider prompt $\mathbf{P}_{\text{eval}}$ ($L_p = 10$ tokens) and executes a single forward pass, yielding logits $\mathbf{z}^{(m)}$ at the final position. Generation is constrained to the binary subspace $\{\text{id}(\texttt{STOP}), \text{id}(\texttt{CONTINUE})\}$ with decision $\hat{y}^{(m)} = \texttt{STOP} \iff z_{\texttt{STOP}} > z_{\texttt{CONTINUE}}$—the natural argmax boundary, requiring no threshold tuning. Because the policy lives entirely in $\mathbf{P}_{\text{eval}}$, interruption criteria are reconfigurable at runtime with zero retraining—unlike trained auxiliary probes (Appendix E).

**No-op path.** If the speaker concludes without interruption, the warm Base cache turns response TTFT from an $O(L)$ cold prefill into a single-step decode. This gain is driven by input streaming itself; Dual-State's contribution is enabling concurrent mid-stream assessment *without* evicting, corrupting, or duplicating that cache.

# 4 Experiments

**Setup.** All experiments run on physical hardware (NVIDIA RTX 4070 Ti, 4-bit precision) with Gemma 4 12B (Gemma Team, 2026), Llama 3.1 8B (Grattafiori et al., 2024), and Qwen 3 4B (Yang et al., 2025). We compare four architectures: **(A) Independent Decider**, a stateless LLM called on every chunk with the full accumulated context, generating free-form reasoning (15–35 tokens) before deciding; **(B) Thought Bypass**, variant of A injecting closed-thought blocks to skip internal reasoning (Gemma only; Appendix A); **(C) Independent + Logit Gating**, stateless full-context re-prefill with a 1-token gated decision, isolating the pure re-prefill penalty; **(D) Dual-State (ours)**, forked streaming cache with a 1-token gated decision.

**Benchmarks.** (i) **FLEXI** (Ge et al., 2025): 400 human dialogue transcripts (200 intentional barge-ins, 200 benign backchannels); (ii) **Full-Duplex-Bench (FDB)** (Lin et al., 2025): 400 dynamic multi-agent roleplay scenarios with semantic ground truth; (iii) **progressive-clue floor control** (Section 4.2).

## 4.1 Interruption Accuracy and Latency

Table 1 summarizes accuracy and latency (full per-model ablations, including thought bypass and confusion matrices, in Appendix A).

**Accuracy is preserved without reasoning tokens.** The reasoning independent decider is accurate (99.75% FLEXI on Gemma) but unusable in real time: its average per-check latency on FDB reaches

Table 2: **HANDRAISER multi-attempt floor control (200 scenarios, $k$=5).** Cumulative redundant context re-prefill tokens per scenario (single-pass baseline: 576.2 tok; Appendix D).

| Model | Architecture | Resolution Acc. | Game Score ($S$) | Time-to-Halt (s) | Consensus Time (s) | Context Re-Prefill |
|---|---|---|---|---|---|---|
| **Gemma 4 12B** | Independent Decider | 49.0% | 6.11 | 0.2437 | 6.6435 | 2661.4 tok |
| | **Dual-State (ours)** | **51.5%** | **6.50** (*+6.4%*) | **0.1492** (*-38.8%*) | **5.6832** (*-14.5%*) | **0.0 tok** (*-100%*) |
| **Llama 3.1 8B** | Independent Decider | 51.5% | 5.93 | 0.1085 | 2.5378 | 3186.5 tok |
| | **Dual-State (ours)** | **54.0%** | **6.59** (*+11.1%*) | **0.0552** (*-49.1%*) | **2.0532** (*-19.1%*) | **0.0 tok** (*-100%*) |
| **Qwen 3 4B** | Independent Decider | 43.0% | 4.92 | 0.0934 | **3.0786** | 3121.9 tok |
| | **Dual-State (ours)** | **44.0%** | **4.97** (*+1.0%*) | **0.0857** (*-8.2%*) | 4.0542 | **0.0 tok** (*-100%*) |

**2,452 ms**. Gated 1-token decisions retain 90.5–100% accuracy across all models, and Dual-State matches or exceeds the gated independent decider (100.0% vs. 90.75% FDB on Gemma).

**Time-to-Halt.** Measured from the arrival of the critical token to the halt command, Dual-State cuts Time-to-Halt from 518 ms to **116 ms** on Llama ($\sim$78%), 1,118 ms to **181 ms** on Gemma, and 191 ms to **86 ms** on Qwen—28–30% below even the 1-token gated independent decider, purely from eliminating re-prefill. This trade-off directly mirrors the Average Penalty Time (APT) objective of semantic interruption detection (Xia et al., 2026).

**No-op TTFT.** When the stream completes without interruption, the preserved cache turns response TTFT into a single decode step: at $L$=1024, TTFT drops by **65.0%** on Llama (234.65 ms $\rightarrow$ 82.18 ms), **48.7%** on Gemma, and **36.0%** on Qwen (full sweep in Appendix B). Fork overhead never exceeds 0.06 ms.

### 4.2 Full-Duplex Floor Control: Progressive-Clue Benchmark

To evaluate end-to-end floor control, we adapt incremental QA (Rodriguez et al., 2019) to a full-duplex protocol (HANDRAISER): a speaker streams progressively more specific clues about a target entity ($k$=5 token chunks, 75 ms intervals); at each boundary the listener may buzz (`STOP`) and guess. Incorrect guesses incur a penalty but the stream resumes, permitting later buzzes. The 200 scenarios comprise 100 adversarial questions from `AdvQA` (Wallace et al., 2019) and 100 from `protobowl-11-13`; a game score rewards early accurate floor-taking (protocol and scoring in Appendix C).

**Results** (Table 2). Under concurrent full-duplex streaming, Dual-State (i) eliminates redundant context re-prefill entirely (the stateless baseline wastes 2,661–3,187 cumulative prefix tokens per scenario); (ii) cuts Time-to-Halt by up to 49.1%; (iii) accelerates consensus by up to 19.1% on Llama (while on Qwen, earlier non-fatal buzzes permit additional clue intake, trading consensus duration for higher resolution accuracy); and (iv) attains strictly higher resolution accuracy and game scores across all three families. In total accounting (Appendix D), the stateless decider consumes 966.3 tokens per 63.4-token scenario (+1424% overage) versus **207.2** for Dual-State (+227%)—a **78.6%** net reduction.

**Ablations.** A chunk-stride sweep ($k \in \{2, 5, 10, 20, 50\}$) exposes a responsiveness–compute Pareto frontier with $k \in [5, 10]$ optimal; trained linear/MLP probes are faster (0.08–0.10 ms) but need task-specific supervision, whereas the prompt-guided Assessor is zero-shot (Appendix E).

## 5 Discussion, Limitations, and Conclusion

**Serving integration & memory.** The fork is a copy-on-write leaf on the RadixAttention (Zheng et al., 2024) trunk, or block-table sharing in PagedAttention (Kwon et al., 2023). Aliasing rather than duplicating the trunk cuts per-stream assessor KV overhead from 90.9 MB to 0.88 MB ($\sim$49.5% at batch 64; Appendix F).

**Limitations.** (i) Our baseline is *stateless*, reflecting decider-as-a-service deployments; a stateful variant would avoid re-prefill at the cost of duplicated ingestion compute and KV memory, and a head-to-head comparison is future work. (ii) Each assessment attends over the full cached history. (iii) Multi-party arbitration, spoken full-duplex integration (Défossez et al., 2024), and human perceptual studies remain open.

**Conclusion.** Dual-State KV-Forking turns a passive streaming listener into an active conversational participant: one persistent cache ingests the stream once; ephemeral forks decide when to take the floor—a reusable building block for real-time full-duplex agent systems.

## References


Amey Agrawal, Nitin Kedia, Ashish Panwar, Jayashree Mohan, Nipun Kwatra, Bhargav S. Gulavani, Alexey Tumanov, and Ramachandran Ramjee. Taming throughput-latency tradeoff in LLM inference with Sarathi-Serve. In *Proceedings of the 18th USENIX Symposium on Operating Systems Design and Implementation (OSDI '24)*, pages 1127–1143, 2024.

Alexandre Défossez, Laurent Mazaré, Manu Orsini, Amélie Royer, Patrick Pérez, Hervé Jégou, Edouard Grave, and Neil Zeghidour. Moshi: a speech-text foundation model for real-time dialogue. *arXiv preprint arXiv:2410.00037*, 2024.

Gemma Team. Gemma 4 technical report. *arXiv preprint arXiv:2607.02770*, 2026.

Yuan Ge, Saihan Chen, Jingqi Xiao, Xiaoqian Liu, Tong Xiao, Yan Xiang, et al. FLEXI: Benchmarking full-duplex human-LLM speech interaction. *arXiv preprint arXiv:2509.22243*, 2025.

In Gim, Guojun Chen, Seung-seob Lee, Nikhil Sarda, Anurag Khandelwal, and Lin Zhong. Prompt Cache: Modular attention reuse for low-latency inference. In *Proceedings of Machine Learning and Systems (MLSys)*, volume 6, pages 341–353, 2024.

Aaron Grattafiori et al. The Llama 3 herd of models. *arXiv preprint arXiv:2407.21783*, 2024.

Woosuk Kwon, Zhuohan Li, Siyuan Zhuang, Ying Sheng, Lianmin Zheng, Cody Hao Yu, Joseph E. Gonzalez, Hao Zhang, and Ion Stoica. Efficient memory management for large language model serving with PagedAttention. In *Proceedings of the 29th ACM Symposium on Operating Systems Principles (SOSP '23)*, pages 611–626, 2023.

Yaniv Leviathan, Matan Kalman, and Yossi Matias. Fast inference from transformers via speculative decoding. In *Proceedings of the 40th International Conference on Machine Learning (ICML '23)*, volume 202 of *PMLR*, pages 19274–19286, 2023.

Guohao Li, Hasan Abed Al Kader Hammoud, Hani Itani, Dmitrii Khizbullin, and Bernard Ghanem. CAMEL: Communicative agents for "mind" exploration of large language model society. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 36, pages 51991–52008, 2023.

Ting-En Lin, Yuchuan Wu, Fei Huang, Luo Si, Jian Sun, and Yongbin Li. Duplex conversation: Towards human-like interaction in spoken dialogue systems. In *Proceedings of the 28th ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD '22)*, pages 3438–3447, 2022.

Guan-Ting Lin, Jiachen Lian, Tingle Li, Qirui Wang, Gopala Anumanchipalli, Alexander H. Liu, and Hung-yi Lee. Full-Duplex-Bench: A benchmark to evaluate full-duplex spoken dialogue models on turn-taking capabilities. In *Proc. IEEE ASRU*, 2025. arXiv:2503.04721.

Tu Anh Nguyen, Eugene Kharitonov, Jade Copet, Yossi Adi, Wei-Ning Hsu, Ali Elkahky, Paden Tomasello, Robin Algayres, Benoît Sagot, Abdelrahman Mohamed, and Emmanuel Dupoux. Generative spoken dialogue language modeling. *Transactions of the Association for Computational Linguistics*, 11:250–266, 2023.

Pedro Rodriguez, Shi Feng, Mohit Iyyer, He He, and Jordan Boyd-Graber. Quizbowl: The case for incremental question answering. *arXiv preprint arXiv:1904.04792*, 2019.

Harvey Sacks, Emanuel A. Schegloff, and Gail Jefferson. A simplest systematics for the organization of turn-taking for conversation. *Language*, 50(4):696–735, 1974.

Gabriel Skantze. Turn-taking in conversational systems and human-robot interaction: A review. *Computer Speech & Language*, 67:101178, 2021.

Eric Wallace, Pedro Rodriguez, Shi Feng, Ikuya Yamada, and Jordan Boyd-Graber. Trick me if you can: Human-in-the-loop generation of adversarial examples for question answering. *Transactions of the Association for Computational Linguistics*, 7:387–401, 2019.

Qingyun Wu, Gagan Bansal, Jieyu Zhang, Yiran Wu, Beibin Li, Erkang Zhu, Li Jiang, Xiaoyun Zhang, Shaokun Zhang, Jiale Liu, Ahmed Hassan Awadallah, Ryen W. White, Doug Burger, and Chi Wang. AutoGen: Enabling next-gen LLM applications via multi-agent conversation. *arXiv preprint arXiv:2308.08155*, 2023.

Kangxiang Xia, Bingshen Mu, Xian Shi, Jin Xu, and Lei Xie. Semantic-aware interruption detection in spoken dialogue systems: Benchmark, metric, and model. *arXiv preprint arXiv:2603.24144*, 2026.

Guangxuan Xiao, Yuandong Tian, Beidi Chen, Song Han, and Mike Lewis. Efficient streaming language models with attention sinks. In *International Conference on Learning Representations (ICLR)*, 2024.

An Yang et al. Qwen3 technical report. *arXiv preprint arXiv:2505.09388*, 2025.

Victor H. Yngve. On getting a word in edgewise. In *Papers from the Sixth Regional Meeting of the Chicago Linguistic Society*, pages 567–578, 1970.

Gyeong-In Yu, Joo Seong Jeong, Geon-Woo Kim, Soojeong Kim, and Byung-Gon Chun. Orca: A distributed serving system for transformer-based generative models. In *Proceedings of the 16th USENIX Symposium on Operating Systems Design and Implementation (OSDI '22)*, pages 521–538, 2022.

Xinrong Zhang, Yingfa Chen, Shengding Hu, Xu Han, Zihang Xu, Yuanwei Xu, Weilin Zhao, Maosong Sun, and Zhiyuan Liu. Beyond the turn-based game: Enabling real-time conversations with duplex models. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing (EMNLP '24)*, pages 11543–11557, 2024.

Lianmin Zheng, Liangsheng Yin, Zhiqiang Xie, Chuyue Sun, Jeff Huang, Cody Hao Yu, Shiyi Cao, Christos Kozyrakis, Ion Stoica, Joseph E. Gonzalez, Clark Barrett, and Ying Sheng. SGLang: Efficient execution of structured language model programs. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 37, 2024.

# A Full Per-Model Ablation Tables

Tables 3–5 report the complete architecture ablations per model, including confusion matrices (TP/TN/FP/FN over 200 positive and 200 negative examples per benchmark), FLEXI latencies, and—for Gemma—the thought-bypass variant, in which synthetic closed-thought blocks (`<|channel>thought\n<channel|>`) are injected to force the model to skip internal reasoning. Thought bypass reduces latency relative to free-form reasoning but remains an order of magnitude slower than logit-gated decisions and is only applicable to models with explicit thought tags.

Table 3: **Architecture ablation: Gemma 4 12B Instruct.** TP/TN/FP/FN reported below each accuracy. Latencies in ms; *Blended* averages over interruption and no-op decisions.

| Metric | Independent (Baseline) | Independent (Thought Bypass) | Independent (Logit Gating) | Dual-State (Proposed) |
|---|---|---|---|---|
| **FLEXI Accuracy** | **99.75%**<br>TP 200/TN 199/FP 1/FN 0 | **99.75%**<br>TP 200/TN 199/FP 1/FN 0 | 98.75%<br>TP 200/TN 195/FP 5/FN 0 | 98.75%<br>TP 200/TN 195/FP 5/FN 0 |
| **FDB Accuracy** | 99.50%<br>TP 198/TN 200/FP 0/FN 2 | **100.00%**<br>TP 200/TN 200/FP 0/FN 0 | 90.75%<br>TP 200/TN 163/FP 37/FN 0 | **100.00%**<br>TP 200/TN 200/FP 0/FN 0 |
| Avg. FLEXI decision latency (ms) | 1063.27 | 823.35 | 260.77 | **95.23** |
| Avg. FDB decision latency (ms) | 2452.00 | 1047.49 | 266.06 | **90.26** |
| Time-to-Halt (FLEXI TP, ms) | 1131.24 | 759.33 | 265.01 | **190.46** |
| Time-to-Halt (FDB TP, ms) | 1117.61 | 746.42 | 268.46 | **180.53** |

Table 4: **Architecture ablation: Llama 3.1 8B Instruct.**

| Metric | Independent (Baseline) | Independent (Logit Gating) | Dual-State (Proposed) |
|---|---|---|---|
| **FLEXI Accuracy** | **93.50%** TP 177/TN 197/FP 3/FN 23 | 91.00% TP 166/TN 198/FP 2/FN 34 | 90.50% TP 162/TN 200/FP 0/FN 38 |
| **FDB Accuracy** | **97.00%** TP 188/TN 200/FP 0/FN 12 | 95.00% TP 180/TN 200/FP 0/FN 20 | 94.25% TP 177/TN 200/FP 0/FN 23 |
| Avg. FLEXI decision latency (ms) | 384.77 | 117.60 | **39.46** |
| Avg. FDB decision latency (ms) | 361.96 | 117.07 | **57.87** |
| Time-to-Halt (FLEXI TP, ms) | 555.19 | 119.79 | **78.91** |
| Time-to-Halt (FDB TP, ms) | 518.45 | 117.67 | **115.73** |

Table 5: **Architecture ablation: Qwen 3 4B Instruct** (4-bit quantization).

| Metric | Independent (Baseline) | Independent (Logit Gating) | Dual-State (Proposed) |
|---|---|---|---|
| **FLEXI Accuracy** | 97.50% TP 198/TN 192/FP 8/FN 2 | 98.75% TP 198/TN 197/FP 3/FN 2 | **99.00%** TP 198/TN 198/FP 2/FN 2 |
| **FDB Accuracy** | **99.75%** TP 199/TN 200/FP 0/FN 1 | **99.75%** TP 199/TN 200/FP 0/FN 1 | 99.50% TP 198/TN 200/FP 0/FN 2 |
| Avg. FLEXI decision latency (ms) | 221.54 | 124.52 | **43.82** |
| Avg. FDB decision latency (ms) | 189.64 | 122.35 | **42.42** |
| Time-to-Halt (FLEXI TP, ms) | 261.13 | 123.57 | **88.52** |
| Time-to-Halt (FDB TP, ms) | 191.19 | 122.63 | **85.70** |

## B No-Op TTFT Benchmark Across Context Lengths

When an incoming utterance concludes without interruption (no-op), the listener must generate its first response token. We benchmark physical TTFT on NVIDIA RTX 4070 Ti in 4-bit precision across context lengths $L \in [64, 1024]$, comparing: **Baseline (cold prefill)**, in which the listener maintains no streaming cache and must prefill all $L$ context tokens before emitting the first token; and **Dual-State / input streaming (warmed)**, in which the Base cache was populated chunk-by-chunk during streaming and TTFT is a single-token decode. TTFT here measures exclusively local GPU latency ($T_{\text{prefill}} + T_{\text{decode},1}$), excluding network and API queuing overhead; full responses require subsequent autoregressive generation ($N_{\text{tokens}} \times$ 15–25 ms). Results are averaged over 3 independent runs per configuration. The primary physical driver of the gain is input streaming itself; Dual-State's contribution is enabling concurrent mid-stream interruption assessment without corrupting or evicting this cache.

## C HANDRAISER Protocol Details and Scoring

**Dataset.** A curated, balanced benchmark of 200 progressive-clue scenarios sampled from official repositories: exactly 100 adversarial human-crafted trivia questions from `qanta-challenge/AdvQA` (of 182 available) and 100 progressive-clue questions from `mgor/protobowl-11-13` (of 3,042 available). Average clue length is 63.4 tokens.

**Streaming turn-taking.** The speaker streams clues in discrete increments of $k = 5$ tokens at 75 ms intervals. Clues begin obscure and become progressively more specific.

**Continuous multi-attempt floor control.** At each chunk boundary, the listener evaluates whether to interrupt (`STOP`) or keep listening (`CONTINUE`). On interruption, it emits a concise entity guess. A correct guess solves and concludes the scenario. An incorrect guess incurs a penalty ($-\beta$) but *the speaker resumes streaming remaining clues*, allowing the listener to buzz again as more discriminating

Table 6: **Physical no-op TTFT across context lengths** (RTX 4070 Ti, 4-bit, 3 runs).

| **Model** | **Architecture** | $L = 64$ | $L = 128$ | $L = 256$ | $L = 512$ | $L = 1024$ |
|---|---|---|---|---|---|---|
| **Qwen 3 4B** | Baseline Cold TTFT | 85.00 ms | 115.08 ms | 100.26 ms | 110.03 ms | 136.47 ms |
| | **Dual-State Warmed TTFT** | **76.87 ms** | **84.95 ms** | **72.15 ms** | **65.34 ms** | **87.28 ms** |
| | *Latency Reduction* | *-9.6%* | *-26.2%* | *-28.0%* | *-40.6%* | **-36.0%** |
| | Measured KV Fork Overhead | 0.01 ms | 0.03 ms | 0.02 ms | 0.03 ms | 0.03 ms |
| **Llama 3.1 8B** | Baseline Cold TTFT | 70.49 ms | 74.75 ms | 88.81 ms | 141.36 ms | 234.65 ms |
| | **Dual-State Warmed TTFT** | **56.12 ms** | **60.16 ms** | **40.81 ms** | **69.14 ms** | **82.18 ms** |
| | *Latency Reduction* | *-20.4%* | *-19.5%* | *-54.0%* | *-51.1%* | **-65.0%** |
| | Measured KV Fork Overhead | 0.03 ms | 0.02 ms | 0.02 ms | 0.03 ms | 0.02 ms |
| **Gemma 4 12B** | Baseline Cold TTFT | 266.93 ms | 242.29 ms | 243.46 ms | 280.28 ms | 447.30 ms |
| | **Dual-State Warmed TTFT** | **228.38 ms** | **237.09 ms** | **162.73 ms** | **229.26 ms** | **229.46 ms** |
| | *Latency Reduction* | *-14.4%* | *-2.1%* | *-33.2%* | *-18.2%* | **-48.7%** |
| | Measured KV Fork Overhead | 0.06 ms | 0.05 ms | 0.05 ms | 0.05 ms | 0.05 ms |

information arrives. If all chunks conclude without a correct mid-stream buzz, the listener makes one final attempt on the complete clue.

**Scoring.** The game score rewards early, accurate floor-taking while penalizing hallucinated premature interruptions:

$$S_i = \mathbb{I}(\text{Solved}_i) \cdot \left[ B + \alpha \cdot \left( \frac{L_{\text{stream},i} - L_{\text{solve},i}}{L_{\text{stream},i}} \right) \right] - \sum_{j=1}^{N_{\text{wrong},i}} \beta \tag{1}$$

where $B = 10$ base points, $\alpha = 10$ maximum speed bonus, $\beta = 0.25$ penalty per incorrect buzz, and $L_{\text{solve}}$ is the token position at which the entity was correctly named.

**Metric disambiguation.** Scenario Resolution Accuracy in Table 2 measures end-to-end task completion on adversarial progressive trivia under multi-attempt dynamics, and should not be conflated with binary interruption precision: on FLEXI/FDB, Dual-State attains 97.25–100% precision in classifying intentional barge-ins, while resolution accuracy (44.0–54.0%) reflects the collaborative problem-solving capability of the listener on adversarially authored questions (Wallace et al., 2019), with Dual-State consistently outperforming the independent baseline by $+1.0$ to $+2.5$ points.

## D Operational Token Accounting

Table 7 reports the complete token expenditure per scenario across the HANDRAISER benchmark (stream length 63.4 tokens, 13.1 chunks of $k = 5$). The stateless independent decider must re-evaluate the accumulating context (576.2 tokens per single pass, escalating beyond 3,000 tokens in multi-attempt games) plus its instruction prompt (130.7 tokens across 13.1 checks) and reasoning output (196.0 tokens), consuming **966.3 total tokens** per scenario ($+1424\%$ overage relative to the 63.4-token stream). Dual-State consumes **207.2 tokens** (63.4 stream + 130.7 assessor prompts + 13.1 gated decision tokens): a **78.6%** net reduction, cutting overage to $+227\%$ ($-84.1\%$). Across 200 scenarios, the stateless baseline wastes over 600K prefill tokens on context re-ingestion alone.

Table 7: **Token accounting breakdown per scenario (HANDRAISER, 200 scenarios).**

| **Component** | **Independent Baseline** | **Dual-State (Proposed)** | **Net Reduction** |
|---|---|---|---|
| Base Stream Ingestion | 63.4 tokens | 63.4 tokens | 0.0% |
| Redundant Context Re-Prefill | 576.2 tokens | **0.0 tokens** | **-100.0%** |
| Decider / Assessor Prompt Tokens ($L_p = 10$) | 130.7 tokens | 130.7 tokens | 0.0% |
| Decider / Output Tokens | 196.0 tokens (Reasoning) | **13.1 tokens** (1-Token Gated) | **-93.3%** |
| **Total Tokens Processed** | **966.3 tokens** | **207.2 tokens** | **-78.6%** |
| **Token Overage** ($\Delta_{\textbf{tokens}} = \mathcal{T}_{\textbf{total}} - L_{\textbf{stream}}$) | **+902.9 tokens (+1424%)** | **+143.8 tokens (+227%)** | **-84.1%** |

## E Ablation Studies: Chunk Stride and Auxiliary Probing

**Parametric chunk-size sweep.** Evaluating floor control at discrete chunk granularities trades kernel-launch frequency against floor-taking responsiveness. Table 8 benchmarks Qwen 3 4B across strides $k \in \{2, 5, 10, 20, 50\}$ on 50 scenarios. At $k = 2$ the agent halts aggressively (10.2 tokens read;

0.6344 s consensus) but triggers 50 forward passes per 100 incoming tokens. At $k = 50$, forward passes drop to 2 per 100 tokens, but coarse quantization lag inflates consensus time to 0.9182 s (32.6 tokens read before halting). Strides $k \in [5, 10]$ form the Pareto knee: sub-720 ms consensus at only 10–20 checks per 100 tokens.

Table 8: **Chunk-size ($k$) ablation (Qwen 3 4B, 50 scenarios).**

| Chunk Stride ($k$) | Checks / 100 Tokens | Time-to-Halt (s) | Consensus Time (s) | Avg Tokens Read | Interruption Rate (%) |
|---|---|---|---|---|---|
| $k = 2$ | 50.0 | 0.1010 | **0.6344** | 10.2 | 94.0% |
| $k = 5$ | 20.0 | 0.1063 | 0.7059 | 20.5 | 72.0% |
| $k = 10$ | 10.0 | 0.1127 | 0.7173 | 28.6 | 44.0% |
| $k = 20$ | 5.0 | 0.1051 | 0.7526 | 31.0 | 44.0% |
| $k = 50$ | 2.0 | 0.1214 | 0.9182 | 32.6 | 62.0% |

**Auxiliary lightweight probing.** A natural alternative to a full assessor pass is to classify interruption from the Base model's hidden state $h_t \in \mathbb{R}^d$ via a linear probe or 2-layer MLP head. Table 9 compares physical execution on GPU. While an MLP head executes in 0.103 ms (vs. 91.50 ms for the full-model assessor pass), auxiliary heads have a decisive limitation for agentic deployments: **zero prompt reconfigurability**. A head is statically trained on fixed offline data; if system instructions, conversational constraints, or few-shot examples change at runtime, the probe requires supervised retraining. Dual-State leverages the foundation model's general reasoning, enabling fully zero-shot, prompt-guided runtime adaptation with zero additional trained parameters.

Table 9: **Auxiliary lightweight probing vs. Dual-State assessor (Qwen 3 4B).**

| Architecture | Trainable Params | Latency (ms) | Zero-Shot Prompt Reconfigurability | Suffix Memory Overhead |
|---|---|---|---|---|
| Linear Probe | 5,122 | **0.084** | No (Requires task supervised training) | **0.01 MB** |
| 2-Layer MLP Head | 656,130 | 0.103 | No (Requires task supervised training) | 1.25 MB |
| **Dual-State Assessor (Ours)** | **0 (Shared)** | 91.50 | **Yes (100% zero-shot, prompt-guided)** | 0.65 MB ($L_p = 10$) |

# F Serving-Engine Mapping, Concurrency, and Safety

**Native mapping to modern inference engines.** Our implementation is benchmarked natively in PyTorch to isolate algorithmic overhead, but the Dual-State mechanism maps directly onto execution-graph primitives of modern serving engines. In SGLang's RadixAttention (Zheng et al., 2024), cached prefixes form a radix tree of physical memory pages: Base ingestion extends the trunk, and each Assessor check is a non-destructive copy-on-write fork instantiating an ephemeral leaf holding only the $L_p = 10$ suffix tokens, freed immediately after the gated decision without copying or altering trunk memory. vLLM's PagedAttention (Kwon et al., 2023) admits the identical pattern through block-table sharing, and modular prompt caching (Gim et al., 2024) can further reuse the fixed decider prompt across checks.

**Multi-stream concurrency and memory scaling.** Under high-concurrency serving ($B$ concurrent dialogues), duplicate KV caches for independent deciders exhaust VRAM rapidly. For an average context of $L = 512$ tokens and assessor suffix $L_p = 10$ on Qwen 3 4B, a duplicate-cache decider consumes 90.9 MB per stream ($B = 1$) and 5.82 GB at $B = 64$. Because Dual-State pointer-aliases the Base cache, Assessor overhead is 0.88 MB at $B = 1$ and 56.2 MB at $B = 64$—a consistent $\sim$49.5% reduction in total KV footprint across batch sizes.

**Emergency trajectory overrides.** Beyond conversational floor control, an agent executing tool actions or generating code can drift into unsafe trajectories. Post-hoc evaluation cannot halt generation before execution completes; the Dual-State Assessor monitors intermediate tokens at chunk boundaries and enables sub-150 ms emergency halts, making it a candidate mechanism for runtime safety supervision.

**Multi-party generalization.** This work evaluates two-party (speaker–listener) turn-taking. Multi-agent swarms require competitive floor arbitration (e.g., token-weighted bidding or priority-tiered interruption), for which per-listener KV-forked assessor branches provide a natural substrate; we leave this to future work.